
Predicting Diabetes Progression

Actionable insights from one-year patient data tracking.



442 Patients



10 Biomarkers

The Primary Risk Drivers

Three standard baseline metrics predict roughly half of future disease severity.



BMI



Blood Pressure



S5 Serum Marker

50%

VARIANCE EXPLAINED

By these three factors alone.

Practical Strategy for Clinics

Translating mathematical models into frontline triage protocols.



Standardize Baselines

Ensure BMI, BP, and S5 are captured at every initial assessment.



Rapid Risk Profiling

Instantly identify patients exhibiting all three high-risk markers.



Prioritize Intervention

Allocate intensive early-intervention resources to this high-risk cohort.

ANALYTIC CONSTRAINTS

Understanding the Uncertainty

The Missing Variables

Our model explains ~50% of progression. The remaining variance relies on unmeasured lifestyle and genetic factors (diet, exercise, compliance).

Population Limits

Findings are based on a specific 442-patient sample. Absolute numerical thresholds may require recalibration for different demographics.



Executive Takeaways

✓ Valid Inferences

Baseline BMI, BP, and S5 are highly reliable early-warning signals for severe diabetes progression over a one-year horizon.

✗ Invalid Assumptions

Do not assume direct causality. Lowering a metric (like BP) pharmacologically will not automatically halt diabetes progression. Data is observational.